

# Muhammad Shiraz Kamran

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## Education

COMSATS University Islamabad (7<sup>th</sup> Semester)  
*Bachelor of Science in Software Engineering* (CGPA: 3.56 / 4.0)

Islamabad, Pakistan  
Expected July 2027

- **Relevant Courses:** Database Systems, Object-Oriented Programming (OOP), Data Structures & Algorithms, Information Security, Software Project Management, Computer Networks, Programming Fundamentals (PF), Software Requirements Engineering, Software Design and Architecture.

## Skills

|                        |                                                                 |
|------------------------|-----------------------------------------------------------------|
| Languages              | Python, JavaScript, Java, SQL                                   |
| Frameworks             | FastAPI, Spring Boot, Express, React, Tailwind CSS              |
| AI & Data              | RAG, supabase, Postgresql, sqlite, AI Integration               |
| Cloud                  | VPS handling, DigitalOcean, Azure, AWS intro, Netlify, Supabase |
| Tools & DevOps         | Git/GitHub, Docker, Tampermonkey, Netlify, Jira, Postman        |
| Certificates & Courses | Cisco Python Essentials 2, Intro to AWS Cloud with Labs         |

## Projects

Mushaf Konnekt | *Flutter, FastAPI, PostgreSQL, SQLite, OAuth2/OIDC, Riverpod, VPS*

Website

- Developed an offline first Quran app as a solo project using **Flutter** and a **FastAPI** backend, integrating the **Quran Foundation Content APIs** and **User APIs** for Quranic content, authentication, bookmarks, collections, and notes.
- Designed the caching and synchronization system using **SQLite** and **PostgreSQL**, allowing users to access Quran data, translations, and saved content offline.
- Implemented authentication using **OAuth2/OIDC** with **PKCE** enabling live synchronization of user bookmarks, collections, and reflective notes.
- Integrated on-device Quran recitation recognition using **fine tuned tiny whisper** (Tarteel) on **Sherpa ONNX** runtime and background **Dart Isolates**, keeping the UI responsive during real-time speech processing and verse matching.
- Used **Riverpod** for state management to handle themes, reader settings, persistent navigation state, and reactive UI updates across the application.

Hadith Semantic Search System (RAG) | *FastAPI, PostgreSQL, React, Docker, Qdrant*

Github · Live Demo

- Developed a semantic Hadith search engine using **FastAPI**, **React**, and **SQLModel**, focused on meaning based retrieval.
- Built a vector search pipeline using **PostgreSQL**, **NumPy**, **FastEmbed**, and the **BAAI/bge-small-en-v1.5** embedding model to generate and store embeddings for similarity search and retrieval through Qdrant Cloud Vector Database.
- Integrated **Google AI Studio** and **Groq** LLM APIs to complete the RAG workflow, and deployed the application on a **DigitalOcean VPS** using **Docker Compose** and CI/CD pipelines with **GitHub Actions**.

Devotol - AI Developer Assistant | *JavaScript, Gemini 2.0 Flash, Netlify*

Website

- Built a lightweight AI assistant using **JavaScript** with a bloat free, minimal architecture to ensure fast load times and quick responses for coding prompts.
- Developed a serverless backend layer using **Netlify Functions** to securely integrate and manage requests to the **Google Gemini 2.0 Flash** model.

## Volunteering

Gaza Relief Fundraising Campaign

Team Co-Lead

- Co-led a team of 5 students to execute a fundraising campaign for Civics and Community Engagement, successfully raising over **350,000 PKR** through community outreach and donation drives.